

HEEL HOOK

Heel hooking is when you pull in with your heel instead of pushing on the hold with your toe, as you would normally. The two most common uses are:

- high footholds, especially on steep terrain
- to improve the direction of force on the handhold you're using.



You can heel hook around a corner in order to shift your weight sideways and maintain balance.

When climbing steep walls you can often place your heel on holds, as long as they are large enough. This will relieve tension and allow you to rest your arms on steep routes and boulder problems. When the foothold is high you can also use the heel hook to pull your body up and close to the wall, instead of standing on your toe. The heel hook can further be used to alter your balance and the direction you're pulling on a hold, for example, when you're heel hooking with your right foot while pulling on a left hand sidepull.



You can use your heel to relieve some tension and rest your arms when climbing steep walls where the foothold is high.

CHOOSING SHOES FOR HEEL HOOKS

In the same way that some shoes are made for tiny edges and others are more suited to smearing, there are also different types of shoes for different types of heel hooks. A round heel with a large surface is better for heel hooks

on larger, more rounded holds, while a narrow heel is better for hooking on edges.



TOE HOOK

Toe hooking is, as the name implies, when you hook your toe around or underneath holds, edges and features. The two most common uses are:

- to improve the direction of force on the handhold you're using
- to keep your feet on the wall on steep terrain.



Synnøve Berg Nesheim uses a toe hook to pull her body rightwards and away from the left hand sidepull.

As with heel hooking, you can use toe hooks to improve the direction of force on the hold you are hanging on. For example, you can use a right foot toe hook to pull your body to the right when you're holding a sidepull with your left hand.

Toe and heel hooks can be used interchangeably, but toe hooks often allow for a longer reach and require less hip flexibility.

By locking your toe underneath holds on overhangs you'll use much less energy keeping your feet on the wall. You can improve the grip of a toe hook by counteracting with your other foot on top of the same or a nearby hold; this is known as bicycling.

In order for a toe hook to work you need to actively use your arms and body to pull your centre of gravity away from the toe hook, and at the same counteract by pulling your body away from the handholds with your toe hook. It is the resulting tension between hands and toe hook that allows us to keep hanging on.

If we lose tension between the holds, the toe hook, the hand, or both, we will most likely slip. Toe hooking requires a great deal of both arm and core strength.



Bicycling.

CHOOSING SHOES FOR TOE HOOKS

Different shoes will work with different types of toe hooks. In general, you want a flat shoe with as much of the upper covered with rubber as possible. Aggressive down-turned shoes often don't have a large enough contact surface. Soft shoes can work well, especially if they are soft enough for the toes to be able to curl up slightly.

DYNO

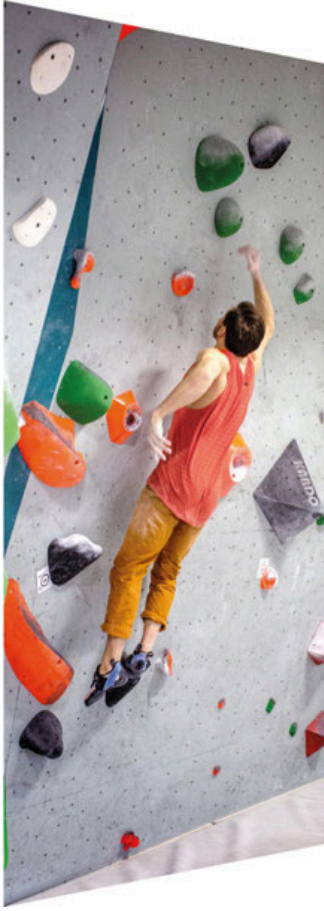
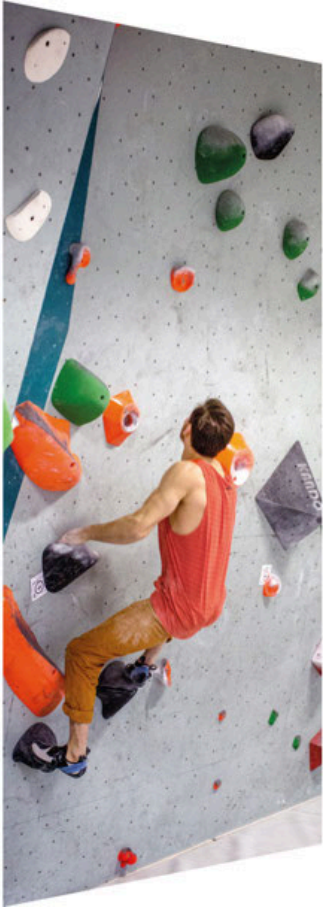
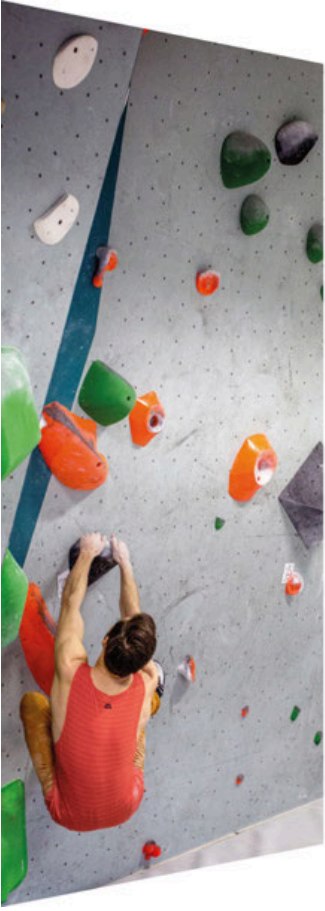
We define a dyno as a move where you're off the wall with at least three points of contact as you go for the next handhold. This technique requires a lot of strength and coordination and many find it mentally challenging.

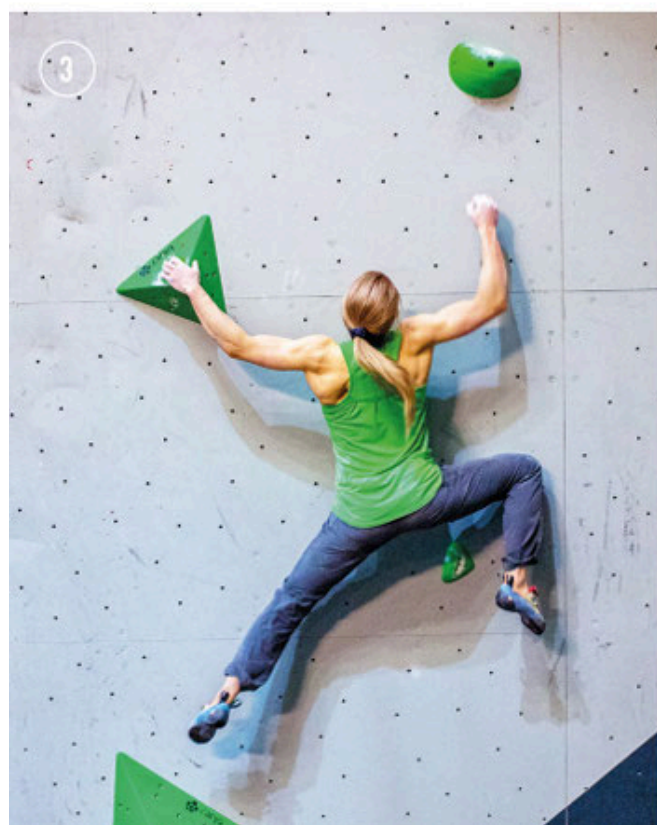
In order to create enough energy to do a big dyno it's important to start the movement in the legs. In practice, this means lowering your body and straightening your arms as much as the position allows for, and then pushing off with your legs. The energy has to be directed towards the next hold, and in a way that keeps your body as close to the wall as possible as you reach the next hold. This is accomplished by using your arms to pull your body close to the wall and steer you in the right direction as you launch with your legs.

It can be physically challenging to hang on after sticking a dyno because of the resulting swing. It's therefore important to have good contact strength in the fingers and arms when doing dynos. Good contact strength will help you pull up as you land the next hold. This can help reduce the swing, which will make it easier to hang on and not lose grip.

Dynos require coordination, and it's important to practise regularly. By practising moves like these, over time your body will learn the nuances which separate success from failure.

Continuously exposing yourself to the mental challenges of dynos and dynamic climbing can build the confidence needed for such moves. Time and time again we've seen climbers who initially struggle with dynos and dynamic moves get better and better as they keep challenging themselves. It's important that coaches and climbing partners do not push too hard, and instead let the climber control the rate of their own progression. Remember that some prefer to climb with less risk involved, while others jump right into things without much thinking.

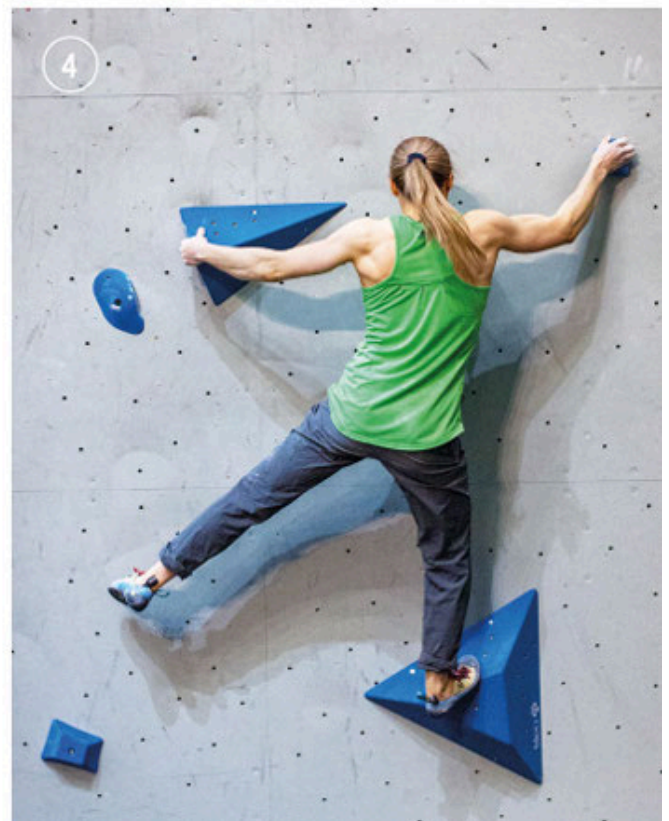
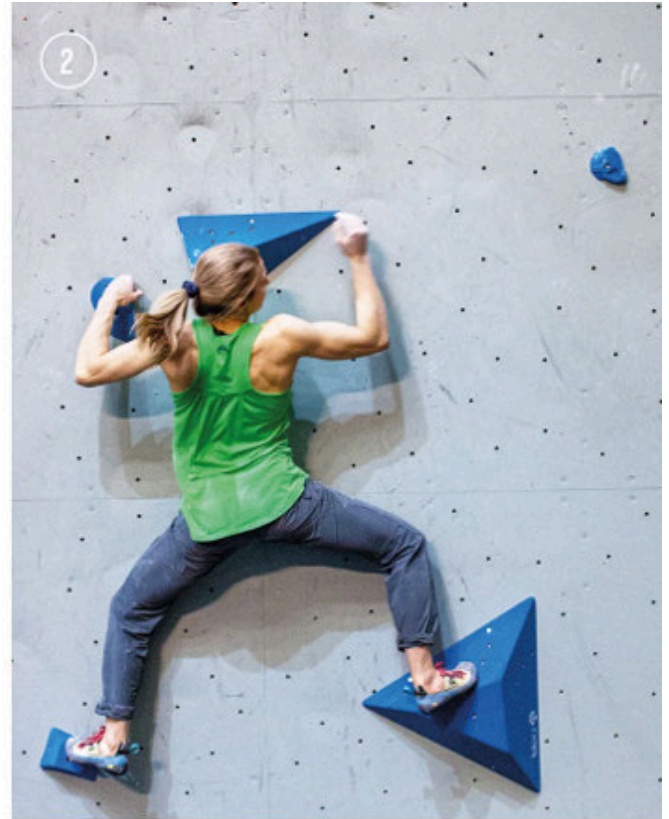




POGO

The pogo – or ninja-kick as it is also known – uses one of your legs as a pendulum to create extra momentum to help you do a dynamic move. By swinging your foot down and behind you before exploding up and forwards, and at the same time pulling your body up and in towards the wall with your arms, you can do big moves from low body positions. And it's in these exact situations where this technique is useful: when the footholds and your body are so low that you can't create enough upwards momentum by doing an ordinary dyno.

The pogo is a complicated technique to master. It requires a great deal of coordination to pull it off correctly, but when done right it can provide the solution to an otherwise impossible move.

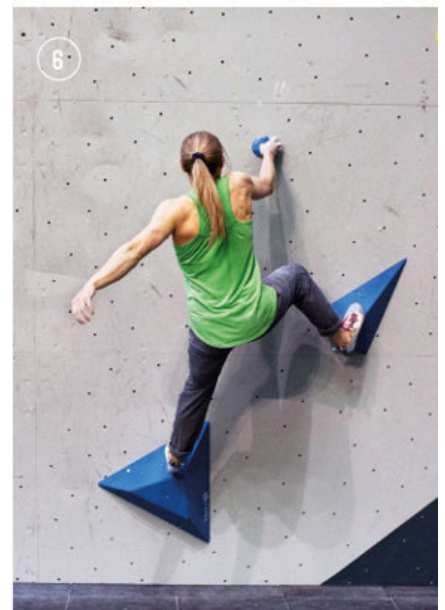
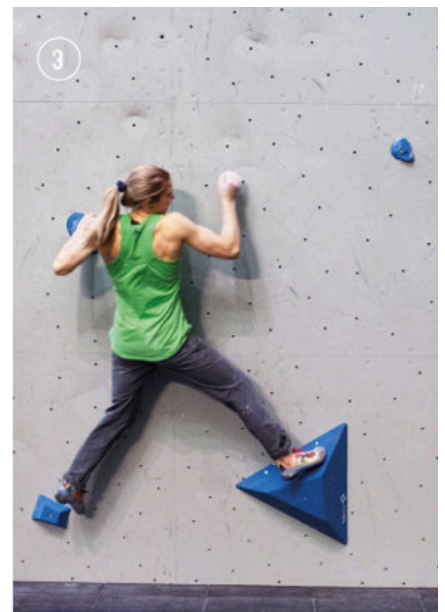


Hannah Midtbø ‘flicks’ her left hand from the starting hold to the pyramid. This way she prevents the barn door effect which would otherwise have swung her off the wall had she only used her right hand.

FLICK, CLUTCH AND MODERN BOULDERING MOVES

Flick and clutch moves have become popular in bouldering competitions as they require a high level of coordination and they look spectacular. The holds have to be placed in exact positions in relation to one other for it to be necessary to do these moves. It's therefore quite rare to encounter moves like this outdoors.

A flick is when you immediately turn your first hand, the one left behind on the starting hold, just as your second hand hits the next hold. This can be necessary when the position your body ends up in as you get to the next hold is impossible to maintain.



An alternative to the flick is to immediately match on the foothold and move your feet as you land the next handhold. As illustrated in the example above, by moving her feet to the next foothold and thereby increasing her area of support, the climber is able to maintain balance as she grabs the handhold with her right hand.

A classic example is when you do a dynamic move to a sidepull and at the same time jump your feet out to a volume to obtain balance as you grab the handhold.



A clutch move is when you immediately continue onwards with either one or both hands to the next hold, if the first hold reached isn't good enough on its own. For example, on a dyno, where the first hold you jump to is too poor to hold, but you continue the momentum up and go for the next and better hold.

Indoor bouldering (especially bouldering competitions) has proven a popular scene for challenging climbers with a combination of moves like the ones described above. A dyno, a clutch and multiple foot matches can all be combined into one single 'move'. The combinations are endless. Modern bouldering competitions are starting to resemble parkour or circus acts, and the 'tricks' described here require a lot of coordination to pull off. You should practise these techniques if you want to be a competition climber, but these skills are rarely needed in outdoor climbing. You should decide what type of climber you want to be, and plan your training accordingly.

WALL ANGLES AND TECHNIQUE



PHOTO: BJØRN HELGE RØNNING

In this section we will describe the fundamentals and techniques that are most useful for different angles of wall. This will be helpful in understanding which techniques will be most useful before you throw yourself at slabs and steep overhangs. We will also provide a useful overview of the different techniques used indoors vs outdoors.



PHOTO: BJØRN HELGE RØNNING

Gina Didriksen stays calm on a tricky slab at Harbak, Norway.

SLABS

'On slabs we are all equal.' This is a well-known saying in the world of climbing, meaning that a big and strong upper body is of no help on slabs.

Slabs are walls with an angle of less than 90 degrees (0 degrees being flat ground). Minute differences in angle on different sections of a wall can greatly impact the climbing experience. Indoor slabs usually have an angle of 87 to 83 degrees, but outdoors you'll find slabs as easy-angled as 70 degrees. This translates to a 'strenuous walk', where your hands are used only for support and balance. When climbing on slabs it's important to lower your heels to maximise the friction between your shoes and the wall. Friction is, as previously mentioned, dependent on the contact surface, and a lowered heel increases the contact surface between your shoe and the wall.



Remember to use soft shoes and to take your time.

Climbing outdoor slabs is mostly about trusting the friction between your shoes and the wall. You have to climb with confidence, and lower your body and poke your rear away from the wall a little in order to achieve optimal friction. Climbing slowly and taking your time to find your balance is usually the best approach. Most times you will want to be fronting, and only rarely is it necessary to twist your hips in. Indoor slabs often feature dynamic moves that require coordination.

VERTICAL CLIMBING

Vertical walls have an angle of exactly 90 degrees. Balance and weight transfer are truly crucial elements, and vertical climbing usually demands a relatively calm and controlled style. This is because the difficulty of vertical climbs is often a result of the size and shape of the holds, requiring us to continuously seek positions that reduce the load on our fingers and arms. Even though most moves will be static, some will require a more dynamic approach. Remember that in these cases you have to be very precise, as the holds can be rather small. More often than not it's better to be fronting on the wall, as long as your flexibility is not a hindrance.

The most common rock type here in Norway is granite. Granite usually involves clean lines and small edges for hands and feet. In order to use such holds efficiently you'll need to climb with precision and use shoes with stiff edges that allow you to stand in a stable position and relax your fingers and hands as much as possible. Other rock types and indoor holds are often less positive, but larger in size. This type of climbing is more dynamic and softer shoes allow you to smear the footholds.



Shift your body weight on to your feet, and be precise when placing your hands and feet.



PHOTO: MIKE HUTTON

Theo Moore going for the on-sight on The Minotaur (E5 6a), Huntsman's Leap, Pembroke, Wales.

OVERHANGS

Overhangs are walls with an angle which is steeper than 90 degrees. The steeper the wall, the more physically demanding the climbing will be; using techniques that lessen the physical impact is essential. Examples include the toe and heel hook, twisting in, and a dynamic climbing style.

The holds are often bigger in overhangs than on vertical walls, and it is usually the distance between the holds that dictates the difficulty of a climb. This favours a dynamic movement pattern. When the climbing gets steeper, twisting in becomes the technique of choice as this reduces the physical impact. Heel hooks and toe hooks are often really useful. A well-placed heel hook can provide a rest for your arms, and both heel and toe hooks can be used to keep your body close to the wall and to better align the direction of force on the holds.

When climbing overhangs it's important to maintain body tension in order to keep pressure on the footholds as you climb. As discussed earlier, we use the term tension to describe the force acting between the different points of contact on the wall. On an overhang it's vital to maintain tension between hands and feet. We often see climbers focus only on the tension between their hands, which results in a loss of pressure on the footholds as they pull their body up. You should instead try to 'pull' your body down using your toes to maintain pressure. To help you maintain this pressure we highly recommend aggressive shoes with down-turned toes. Some shoes are so soft that you can curl your toes behind the hold where possible. Legendary American climber Dave Graham says he regards his feet as claws, gripping the footholds to keep his feet on the wall.



Twist in, maintain pressure through your feet and move swiftly!



PHOTO: MIKE HUTTON

Tom Randall on The Spider (F8a), Chee Dale, Peak District, England.



PHOTO: BJØRN SÆTNAN

Mari Augusta Salvesen dispatches The Rhino (Font 7b+), Rocklands, South Africa.

ROOFS

Roof climbing is, as the name implies, climbing on walls with an angle that is near horizontal (180 degrees). This is the most physically demanding type of climbing, but luckily there are some tricks you can use. Just like when climbing overhangs, there's less of a need for techniques that focus on balance. The focus is instead on dynamic climbing and trying to reduce the load on your fingers and arms.

When climbing roofs you should actively use toe and heel hooks to unload fingers and to avoid big swings that will inevitably occur if you lose tension on the footholds. Bicycling footholds is especially common to keep your feet on the wall and to make the climbing less physical. Twisting in is, for the same reason, the technique of choice.

As with overhangs, soft and aggressive shoes are the best for climbing roofs as they allow you to curl your toes behind footholds, thus helping you keep your feet on the wall.



Use any trick in the book to keep your feet on the wall!

‘LEARN TO DO BY KNOWING AND TO KNOW BY DOING.’

JOHN DEWEY



TECHNIQUE TRAINING

In a way, we are all training our technique every time we go climbing. Every time we put on our climbing shoes and move around on the wall we're practising different techniques and constantly assessing whether to move dynamically or statically, and how to adjust our balance. But you should still set aside some sessions to focus specifically on certain techniques to ensure the quality of the training. Such sessions should follow these guidelines:

- You should be well rested, motivated and curious.

- You should define focus areas in advance.

- During the session, maintain focus:

- How should the move be executed?

- How am I executing the move?

- How can I improve the execution?

The first point is perhaps the most simple. Yet we still see many climbers fall victim of the temptation to train hard and physical when they feel fit and rested. Technique can be trained on its own or as part of a climbing session, but it should never immediately follow a physically demanding session as you will be less focused. Technique training is about discovering new movement patterns and nuances of your already established techniques. You need to be fresh, motivated and curious to challenge the status quo, and be open to new ideas from your peers.

For technique training to be effective you should define one or more focus areas. These should be based on your weaknesses as a climber. Some climbers are well aware of their strengths and weaknesses, while others are not. Those of you who are unaware should seek help from either a close friend or an experienced coach who can point out areas you should focus on. It's important that the person giving you this feedback knows you as a climber and spends some climbing time with you. An experienced coach will, within a few hours of varied and challenging climbing, be able to point out your strengths and weaknesses. Most larger commercial climbing walls will have experienced coaches available to help you with such an evaluation. Alternatively, you can test yourself on different boulder problems and routes and see if you can spot a pattern and assess which types of problems or routes you find easy or hard.

One of the biggest challenges of identifying strengths and weaknesses is that it's often possible to climb a boulder or route in the style that suits you the most. For example, a static climber will often be able to do a move statically, even if this move should ideally be performed dynamically. Being made aware of this makes it possible for the climber to plan sessions where the focus is primarily on dynamic climbing. By starting a process where they continually challenge themselves in a structured manner, dynamic moves will over time become a part of their normal movement pattern. This way they're not only practising dynamic moves when specifically training them, but also when they're 'just' climbing.

When training specific moves and techniques, it's important to have a clear picture of how a move should be executed and how you're actually doing it. Try paying really close attention to how other climbers are doing it, especially those who are doing it correctly. To understand how you yourself are doing a move it's helpful to discuss with others and to film yourself climbing. Having an experienced climber or coach point out strengths and faults in your execution at the same time as watching a video of yourself climbing is a powerful tool in understanding and developing your technique. Remember, it takes a trained eye to spot the minute details of climbing techniques, and whoever you're discussing it with should be experienced in coaching climbers and know your strengths and weaknesses. At the same time you need to be open to new ideas and spend time in the ongoing process.

STAGES OF DEVELOPING TECHNIQUE

Learning a new technique takes time and effort. When describing the progression of learning a certain technique, it is common to look at technique training as a repetitive exercise where you go through different stages:

FAMILIARISATION

Here you get an overview of how the technique is performed. This usually starts with watching other climbers before trying it yourself. When you try to perform the technique yourself it can be a good idea to divide the technique into familiarisation exercises which contain elements of it. This will make it easier to combine all the different elements into one movement. Using deadpointing as an example, the familiarisation exercises could be:

- finding the starting position for the move
- deciding the direction of the movement
- defining the optimal position on the wall at the end of the move.

Becoming comfortable with these three elements will make it easier to start trying the whole movement and to enter the next stage.

COARSE COORDINATION

Here you practise the moves until you can consistently repeat them. At this stage it's reasonable to pick three to five boulder problems or challenging individual moves that require the chosen technique for this training session. Get help from a coach or an experienced climber who can define the moves, or find a suitable boulder problem at your local climbing wall. It's important to be conscious of what makes a move harder or easier, and to get feedback from others on your execution. Filming yourself climbing is an important tool at this stage.

FINE COORDINATION

Here you focus on the details of the movement. It's important to have a high number of repetitions and to really fine tune the moves until they are executed to perfection. At this stage the feedback you get from your own body is more important than what others tell you. This is because it's difficult for others to pinpoint the details that

make something feel right or wrong for you individually. Filming yourself in slow motion can help give you an even better understanding.

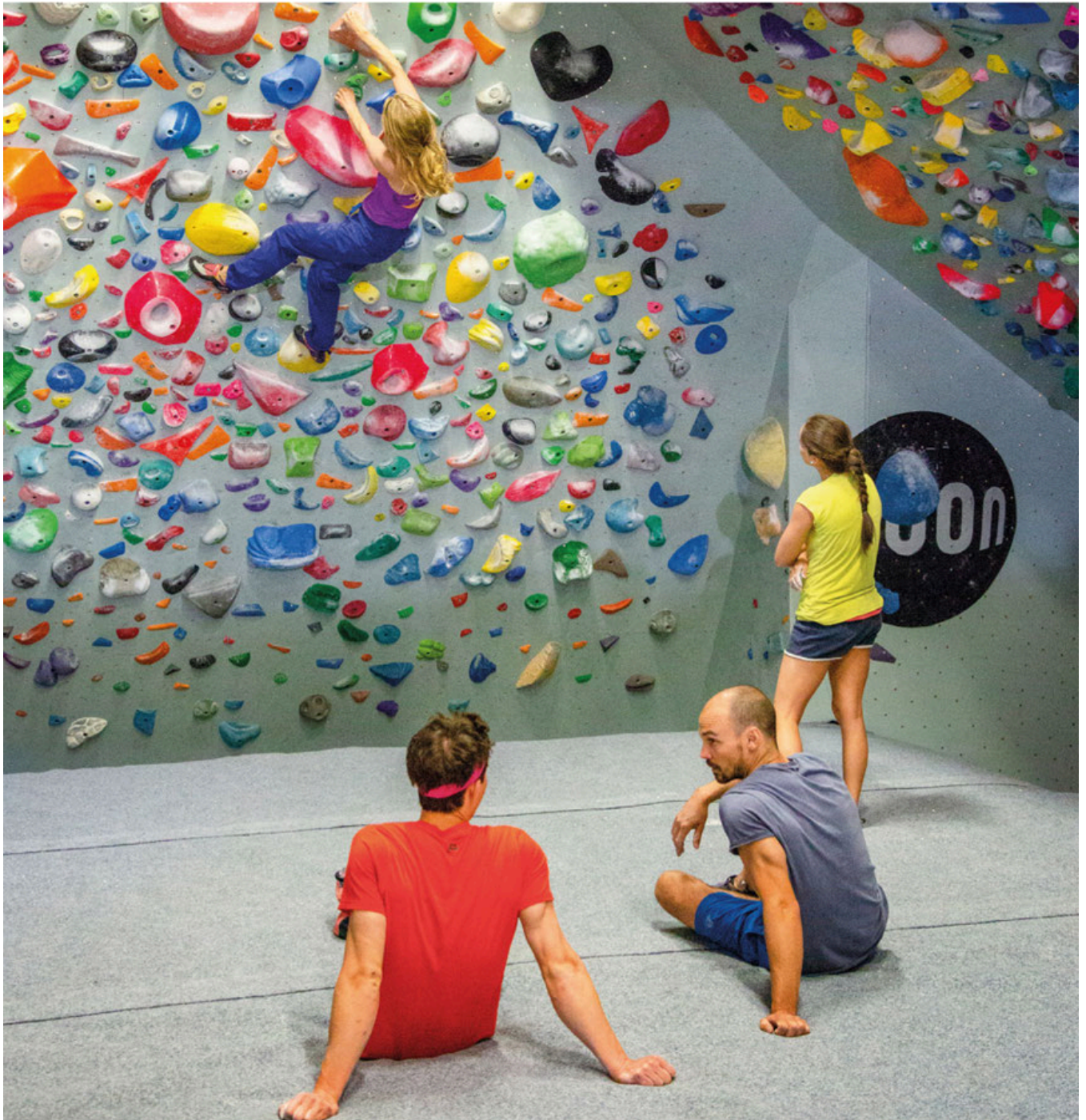
AUTOMATION

At this stage your body should be on autopilot and you should be climbing without thinking about the move itself – as long as the conditions are similar. But just because a move is automatic doesn't necessarily mean it's also optimised. Most moves have some scope for variation, but only one way is the optimal way. This means that even though you have learned a new technique, it's important to stay curious in order to optimise it. Getting feedback from others, filming yourself and listening to your body when climbing are the most important ways to improve.

ADAPTATION

This is perhaps the biggest challenge for us as climbers. There will always be small differences between different wall angles, formations, and types and sizes of holds. Even if a certain technique is automatic on a particular boulder problem or route, you will still have to adapt it when you come across similar moves. Experienced climbers often look like they're climbing purely on intuition. This is a result of them mastering different techniques and not having to stop and think every time they're about to do a move. They recognise and initiate the moves automatically.

External factors also play a role. For example, it can be quite difficult to climb with perfect technique during a competition with lots of spectators, or when you're run out far above the last piece of gear or bolt on a route. External factors will always affect your climbing. You should be prepared for this and minimise the effect it has on your technical execution.



There are large differences in the complexity of different techniques. More complex techniques require a lot more coordination and demand higher precision. These techniques require many familiarisation exercises and much training before they can be mastered. They will also be difficult to reproduce in new situations. Some techniques might only be suited to one single move on one single boulder problem. Others can be useful just about all the time.

Beginners and intermediate climbers should therefore focus on the techniques they will need the most often – those regarding balance, and simple dynamic techniques.

Remember that technique training is like anything else: practice makes perfect. If you want to become better at lead climbing on vertical walls, you need to practise the techniques needed for that style of climbing. If, on the

other hand, you want to become a competition climber you should focus on coordination moves and dynamic climbing.

LEAD VS BOULDERING

Lead climbing is all about finding your flow and using the minimum amount of energy so you can hold on for as long as possible. In bouldering, the difficulty often lies in doing just a single move, challenging your strength, coordination or balance to the limit. This begs the question: should you train for both lead and bouldering if you're mostly going to do one or the other? If you ask us, the answer is yes. Climbing technique is complex: there's always something to learn from another branch of climbing.

Despite there being short bouldery routes and long boulder problems, there are several reasons that those seeking progression should train both lead climbing and bouldering and focus on multiple facets of technique training – those associated with bouldering and those associated with lead climbing. This is regardless of whether you want to excel at one or both genres, because some techniques are easier to learn when climbing on a rope, and others are easier to learn when bouldering.

In general, we recommend that you do more of the specific technique training when you're bouldering. It's easier to design specifically themed challenges on the bouldering wall than on the rope wall. There is also less of a mental aspect, which is especially important when training dynamic techniques. By practising these techniques when bouldering you'll become comfortable with them and over time include them as a natural part of your repertoire, enabling you to do difficult and challenging moves when roped up. Yet there are some techniques that are easier to understand when climbing on a rope – especially techniques involving weight transfer and balance. This is a result of your arms getting tired and being unable to pull hard, which forces you to lower your body, find the balance and shift as much of your weight as possible on to your feet. Way too often we see boulderers fail to master these techniques. Their tendency is to grip harder than they need to and overuse their arms. This results in loss of pressure on the footholds and an increased risk of falling. By doing more roped climbing, boulderers can learn the importance of balance and energy-efficient climbing – also significant aspects when bouldering.



PHOTO: BJØRNAR SMESTAD

We must also not forget that maybe the most exciting – and also the most challenging – part of climbing is the fact that we never practise the exact same move. There's always a level of improvisation in the movements we do as we have not done this exact move before. This means we need a large repertoire of moves to choose from, to ease the adaptation and improvisation needed for doing new moves. By climbing as many routes and boulder problems as possible, we are constantly adding to our movement library and making it easier to improvise every time we try something new. The more routes and boulder problems we climb, the better.

Kilian Fischhuber, one of the most decorated competition boulderers of all time, claimed he was never the strongest or most technical climber. His biggest advantage was that he was extremely skilled at regulating his stress level. He always knew exactly how much he could relax in a given position, or how hard he had to pull in another. He conserved his strength where he could, and used just the right amount of power when he needed to explode off a hold. He only competed in bouldering, but much of his training was done on the rope wall.



CLIPPING THE ROPE

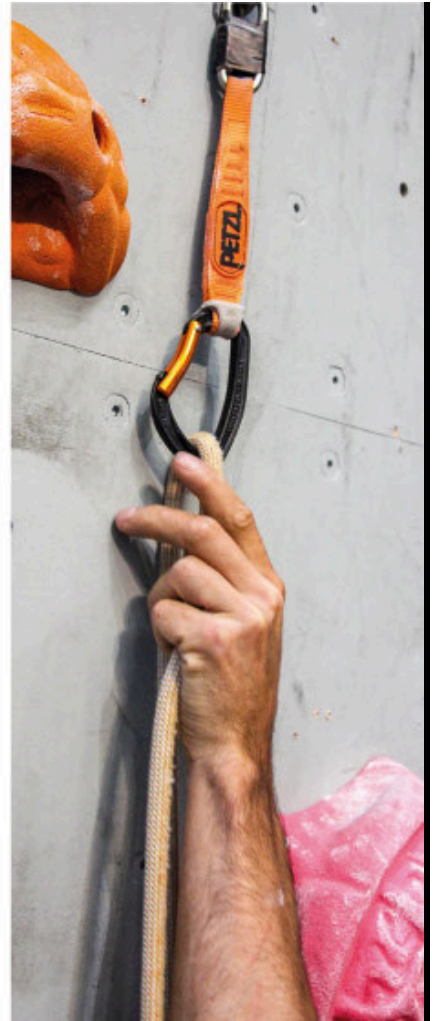
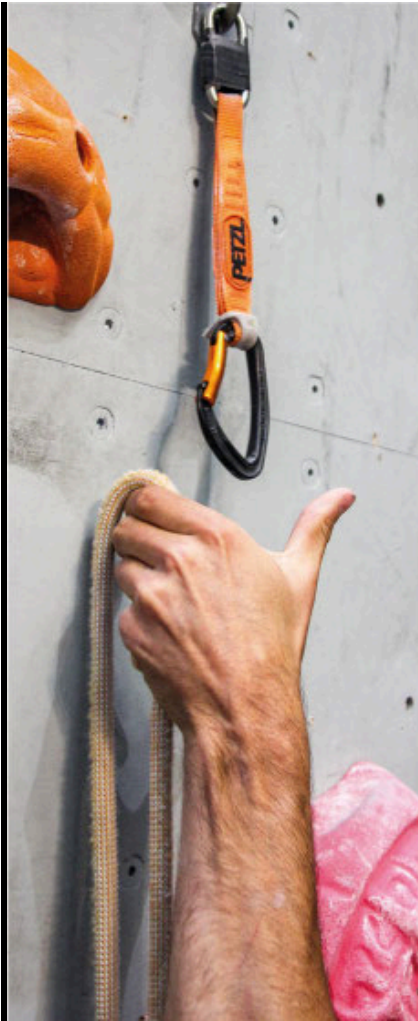
Anybody who has been through the process of learning to clip the rope into quickdraws knows how frustrating this process can be. Not only are you now standing far above the last bolt, but you've also pulled up an extra length of rope to reach the karabiner, which could potentially result in a long fall. You start to fumble with the 'biner, the rope is twisting away and you can't get it to slip into the gate (the part that opens). You're standing slightly off balance, and the arm that's keeping you from falling is getting tired. It feels like your foot is starting to slip and you're just fumbling even more with the quickdraw. Sound familiar?

For an ascent to be considered valid it has to be done on lead, and having an efficient way of clipping the rope into the karabiner is an essential part of whether you succeed or not. As with learning any technique, clipping must also be practised. Start in safe and controlled situations: there's nothing wrong with standing on the ground and practising clipping a low quickdraw over and over until you get the hang of it. Continue by moving on to lead climbing an easy route, where you have enough margin to take your time with every clip. As you keep practising you'll progress naturally and become more comfortable with climbing harder lead climbs, clipping from smaller holds and more strenuous positions, and with an increasing level of pump in your forearms. On really hard routes, clipping the rope can be the hardest part of the climb, and being as efficient as possible when clipping is crucial.

Below are the most common techniques for clipping the rope. Experiment with all of them to find the one that works best for you and then optimise it; but we recommend you learn all the different techniques well enough so that you can adapt your choice based on what you encounter on different routes.

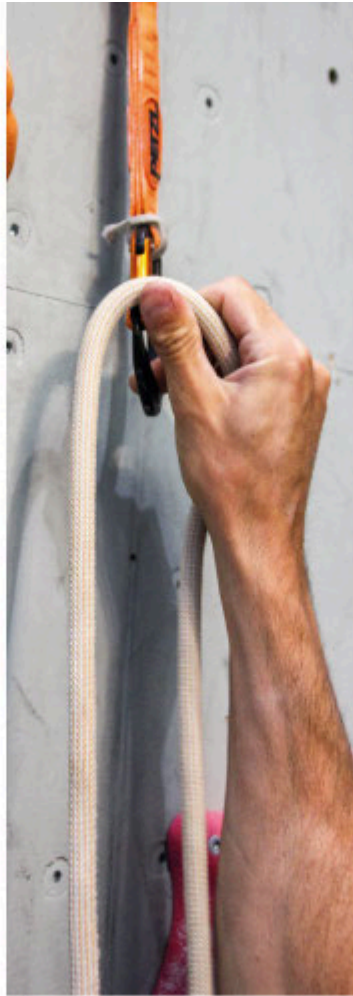
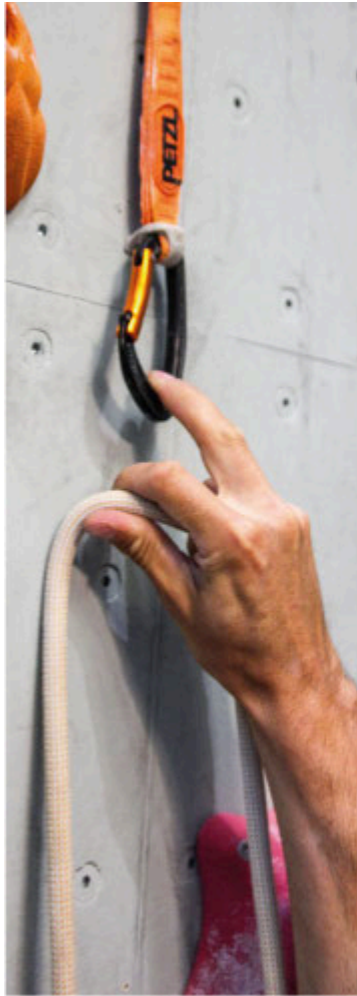
FOREHAND

When the gate of the karabiner is facing the same side as the hand you're clipping the rope with (for example, the gate is facing left and you are clipping with your left hand, as illustrated in the photos, right), it's efficient to let the rope run from your harness, up through your hand and over your index finger. This way you can hold the karabiner with your thumb, open the gate with your index finger and the rope, and drop the rope through the opening.

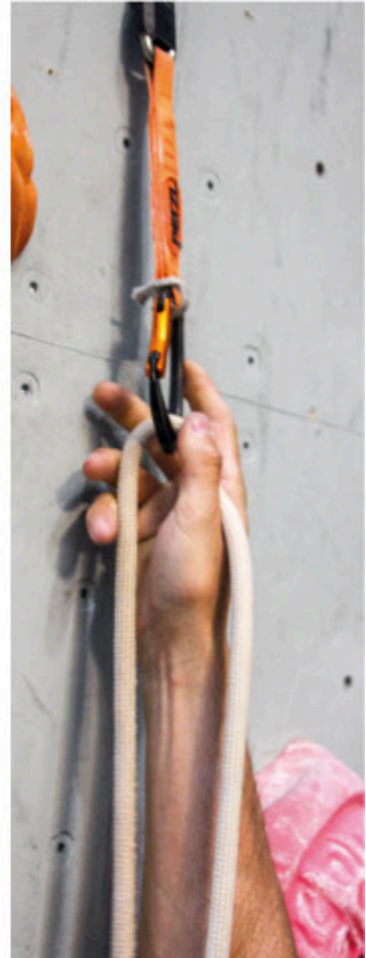
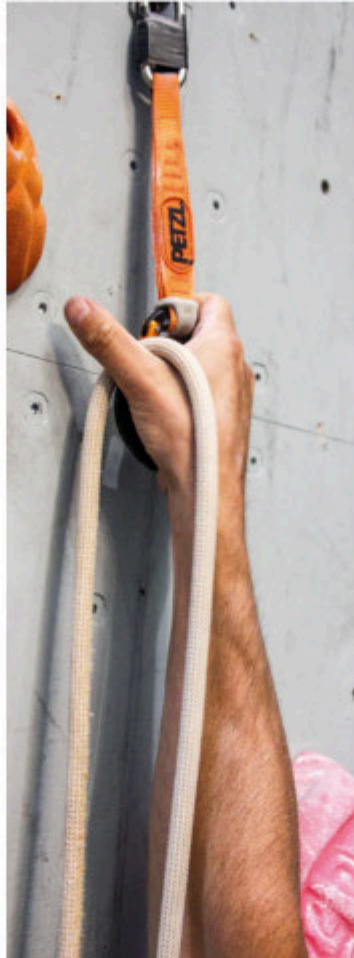


BACKHAND

When the gate of the karabiner is facing away from the hand you're using to clip with (for example, the gate is facing left and you are clipping with your right hand, as illustrated in the photos), it's common to use one of two techniques:



1. Hold the rope between your thumb and index finger. Hold the karabiner steady by pulling down on it with the tip of your middle finger. Use your thumb and index finger to guide the rope and push it through the gate with your thumb. This is an effective technique for slabs and vertical climbing, as you can use your middle finger to lift the karabiner away from the wall slightly to add some space for the rope between the karabiner and the wall. It's also useful when clipping up high or reaching out to the side, as you can maximise reach by using your middle finger to reach the karabiner. Keeping the karabiner steady can be challenging, particularly on steep terrain where you can't rest your hand against the wall behind the karabiner.



2. Let the rope run from your harness and over your palm between your thumb and index finger. Grab the quickdraw directly at the lower karabiner with your index and middle fingers to hold it steady. Twist the karabiner into your palm and use the meaty part of your thumb to squeeze the rope through the gate. This is an efficient technique when climbing on steep terrain where the quickdraw is hanging freely below the overhanging wall, as it helps you hold the karabiner steady. You will lose a bit of reach though when compared to the first technique, above, and it's therefore not as efficient when you have to reach up high or out to the side.



PHOTO: BJØRNAR SMESTAD

READING ROUTES AND BOULDER PROBLEMS

When you're visualising, you can picture how to climb a route, make a plan and execute it. This gives you a good understanding of technique. Discuss the plan with your peers to get their input, as other climbers might have completely different ideas on how to solve a move or section of climbing. It's important to imagine how to use both handholds and footholds, where to move fast, and where to move with more control. It can be hard to read the whole route when outdoors, but indoors you can usually see all of the holds. When reading routes indoors it can be a good idea to put yourself in the route setter's shoes. If there's a foothold out left, it's probably there for a reason. Route setters don't put holds up at random in the hope that someone will find a way up; they plan for a certain sequence and to force the climber into specific positions. By picturing how your body must be placed in relation to a given set of handholds and footholds, you're making a more conscious decision about how to climb than if you were climbing purely by intuition. This will, in turn, improve your intuitive climbing. Visualisation is important to improve your technical abilities, but it's also an important mental tool. We'll dive deeper into this in the chapter on mental performance factors.

What happens if you read a climb incorrectly? Unfortunately, this happens quite often, and it can be hard to reset and improvise when the challenge you face is completely different from the one you prepared for. You have to think quickly so as not to get tired, and go for a new solution. Sometimes a new solution will present itself, and sometimes you have to stop and think before continuing the climb. Some climbers train specifically for these situations. They have sessions where they don't read a route before starting to climb, forcing the whole route to be climbed by improvisation. This can also be trained when climbing on sight outdoors, as you can see less of the holds from the ground which makes reading the whole route difficult or even impossible.

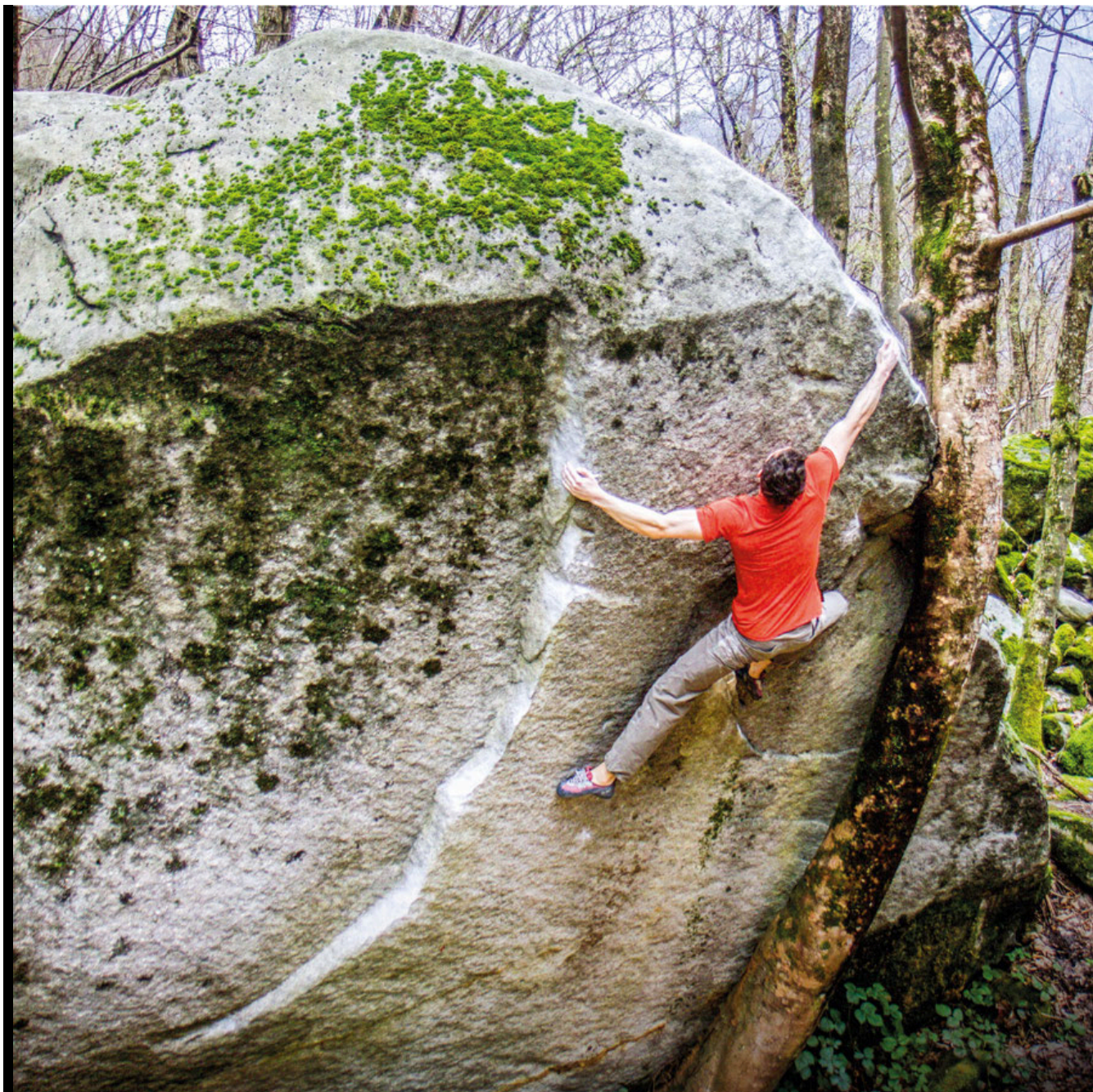


PHOTO: THILO SCHRØTER

Stian Christophersen flashing Special Edition (Font 7c+), Brione, Switzerland.



SET YOUR OWN ROUTES AND BOULDER PROBLEMS

THE DEVELOPMENT OF larger commercial climbing walls has greatly reduced the need to invent your own routes and boulder problems. These gyms have their own professional route setters who work to provide you with different routes for training anything from strength to endurance to technique. This is, of course, a good thing, but making your own routes and boulder problems is an important part of becoming a better climber because it forces you to become more conscious of why you move a certain way when climbing. By setting your own problems you're guaranteed to also become better at solving them, which is such an important part of climbing, and you'll be able to set problems with the perfect level of difficulty and technical focus for the exact training session you're about to have. When to start setting your own problems is dependent on your own development as a climber and on your local climbing wall. If your climbing wall is rather small, with just a small bouldering wall with lots of random, unmarked holds, it will be natural for you to start setting your own problems earlier than if you normally go to a large commercial climbing wall.

You'll need to set boulder problems with specific technical themes. If, for example, you're going to challenge yourself with a deadpoint move, you need to consider which holds to use and where to place them in relation to one other to force the deadpoint move. You will also need to set problems and circuits to ensure the quality of your physical training. To complete a bouldering session which targets finger strength and static strength, you need good quality problems on small holds suited to your strength and height.



We have on multiple occasions observed the Kolsås Climbing Club youth group training here in Norway. The kids are encouraged to set problems for themselves and each other at an early stage. Over time they develop their skills and achieve a good understanding of climbing. They intuitively understand how to move on the wall, and how to solve a problem by looking at it.



Setting your own problems is harder than you'd think. We asked one of Norway's top route setters, Kenneth Elvegård, to give us some tips:

- Set your own variants of existing problems. You can subtract or add holds to change the character or adjust the difficulty level. Many shorter climbers can, for example, add an extra foothold to make a height-dependent problem possible.
- Set your own problems on a wall which is littered with holds. The classic training wall is normally packed with holds. This gives you endless combinations to choose from and is a good place to start.
- Start by choosing which handholds to use.
- You can choose to use all holds for feet, just a selection, any tiny foot jib, only the same holds as you used as handholds, or any variation thereof. Using only the same holds for feet as for hands is called trailing, and is suitable for problems that don't traverse sideways too much. If you're traversing you might have to include some extra footholds or jibs. Jibs are small footholds set on the wall to be used solely as footholds. They are usually too small to be used as handholds and are often attached using screws instead of bolts.
- Decide up front what you want to set, and don't give up until you've accomplished your goal. You can start by setting just one move, and then take it from there. Remember to keep testing each move. Route setting consists of setting, testing, tweaking and retesting.

When training your technique, you should choose a wall with the appropriate steepness for the technique that you're going to focus on. For precise footwork, and balancy and static climbing it's better to choose a vertical wall, while overhanging walls are better for dynamic and flowy moves. Tension and directing force is needed on any wall angle, but more so on steep prowls, where you compress together with both hands and feet, or on steep overhangs with poor holds, where tension is needed from head to toe. Here are some tips:

- To train weight transfer you should use large protruding footholds, and small, preferably poor, handholds. Set a problem that traverses sideways slightly, so as to feel the subtle differences of placing your hips slightly more to the left or right.
- Precision footwork is best trained on small footholds or footholds that require you to place your foot in exactly the right spot. You can use slightly better handholds, but not so good that you can afford for your feet to slip.
- Static climbing is best trained on small holds that are difficult to stick at speed, such as small edges or pockets. Footholds that are large and positive are well suited, as they allow you to pull with your arms without your feet slipping because of reduced friction.

- When training dynamic moves you should either make the moves longer and/or the holds worse. Longer moves are harder to do statically, and slopey holds get worse as you pull yourself higher. Both force you to start the movement from a low position. Try using volumes for feet. If these aren't too positive, they will often force a dynamic move.

- For directing force and maintaining tension you need holds that are too poor to hold individually. Try experimenting by holding the smallest holds possible on each side of a prow. By exploiting the opposing forces you generate, you can use much smaller holds than you might think. When climbing an overhang, try also using holds that are so small that you'll fall if your foot slips. You can start with a larger hold, and work your way down to smaller footholds as you improve.

Physical routes and boulder problems are easier to set than technical ones, as they don't require the same level of precision. Here are some tips to help you set physical problems:

- Try to avoid dirty tricks when setting physical problems. We recommend fronting as much as possible, and no twisting, heel hooking or matching on too many holds. The point of physical training is to train your physique, and using advanced techniques that reduce the load will defeat this purpose.

- If you're working your finger strength on a five-move boulder problem, you should use holds that are so bad that you can only manage to do just those five moves. If, instead, you want to do a 50-move endurance circuit, you need to use holds that will enable you to do those 50 moves.

- For arm and upper body strength training, you should use larger holds and do bigger moves. It's OK to cut loose with your feet now and then, but try to avoid excessive flailing.

- For finger strength, you obviously need small handholds, but you can vary the hold size somewhat, by dynoing to slightly larger holds and going static between smaller holds without cutting your feet.

- Always test the boulders before you start training to ensure that they fit your programme.

OTHER PEOPLE'S PROBLEMS SHOULD BE YOUR PROBLEMS TOO

Most of us have a tendency to only climb boulder problems that suit our style. By trying problems which other climbers have set, and preferably climbers with a different style than your own, you'll also be training your weaknesses. This is a good way to become more conscious of your own strengths and weaknesses. If parts of a boulder problem are impossible because of height/reach or the like, try adding a foothold or changing that part of the problem.



HANNAH MIDTBØ:

Progression takes time and effort. If you're investing time and effort, make sure your training is varied.

INDOORS VS OUTDOORS

Too many climbers favour either indoor or outdoor climbing, without reflecting on the elements that they're missing out on by not including all forms of climbing in their training. Sure, a competition climber should focus their training indoors, just as a trad climber loyal to outdoor cracks might not benefit too much from jumping around on volumes indoors, but as we've mentioned previously, climbing is about using the techniques we have in our arsenal to improvise in new situations. We should therefore always seek to challenge and increase our repertoire as much as possible.

An indoor climber will benefit from climbing outdoors because:

- outdoor climbing quite often involves moving around on smaller and poorer holds for both hands and feet. This requires precision and sensitive climbing, which is hard to replicate and train indoors. Many climbers who only train indoors don't have a good understanding of weight transfer and keeping a low centre of gravity, as the holds indoors are usually not conducive to such types of moves.
- outdoor climbing can sometimes involve painful and awkward holds and moves, which route setters usually try to avoid with indoor climbing. If we recognise the value and importance of having a large repertoire of moves, we can't deny ourselves the opportunity to learn something new, just because it feels uncomfortable or because the holds take too much skin.

An outdoor climber will benefit from climbing indoors because:

- indoor climbing, especially bouldering, makes it easy to train specific techniques. Whether it is routes set by others, or a problem where you define what holds to use yourself, it will be easier to try one or a combination of moves repeatedly and evaluate the process.
- indoor climbing is often more dynamic and requires more tension, which is difficult to adequately train outdoors.



PHOTO: BJØRN HELGE RØNNING

BETA OR NOT?

Most climbers today will have seen another climber on their route or boulder problem before they try it, or they will have received some information on how to do it. This information is known as 'beta'. Having beta can be both good and bad for developing your technique. Beta is positive in that it can serve as inspiration for trying new boulder problems and new techniques. But it can also have a negative effect as it simplifies the process of problem-solving. Climbing is problem-solving, and you won't improve if you don't actually practise solving problems.

We didn't use to have the enormous database of online climbing videos that we have today. When climbers went out to try a problem, they usually had to figure things out themselves. This turned them into inventive and resourceful problem-solvers. Many climbers today lack this ability and perhaps also the motivation needed to solve these problems. They want to get to the top as quickly and easily as possible, which of course makes beta invaluable. This takes some of the magic away from climbing. Part of what makes climbing so exciting and unique is that we don't practise predefined moves, but use our own body, skill and experience to solve these problems.



Challenge yourself by hitting the climbing wall and trying as many problems as you can without watching anybody else try them before you. Experience the joy and sense of achievement from solving problems on your own. For us, this process is always rewarding, and we're convinced it has improved our technique.

During one of our trips to Rocklands in South Africa, we had a chat on this subject with one of the best competition climbers of all time, Kilian Fischhuber. We wanted to convince Kilian to come to Norway for some climbing and were wondering if he had seen any of the videos we had made. He responded that he almost never watched climbing videos. This piqued our interest, as a world without climbing movies sounded terribly sad to us, so we asked him why. He said he didn't want to know how a boulder problem should be solved, as this took away from his interest in doing it. To Kilian, climbing was solely about problem-solving.

SUMMARY

In this chapter we've presented fundamentals and techniques we feel are universal and useful for any and all climbers. Yet there will always be individual differences, and your physical abilities and attributes will affect which techniques will suit you better. This means that the way you and your climbing partner solve the same boulder problem or route might be drastically different, without one way necessarily being any more correct than the other.

A common example of this is the difference between slender and powerful climbers. Slender climbers usually have quite strong fingers relative to their body weight. They will, therefore, be able to do moves on smaller holds which heavier climbers cannot. A larger climber might prefer to do bigger moves on bigger holds. The lighter climber will over time develop the skill to keep their feet on the wall because they don't have the upper body strength to hold on and stop a swing. One of the world's most well-known climbers, Dave Graham, belongs to this category. He has said he thinks of his toes as claws that keep his feet latched on to holds. If his feet slip, he falls. 'Skeleton style', as he himself calls it, works well for him, and there are very few routes or boulder problems he cannot climb. Chris Sharma, on the other hand, is much more of a powerful type of climber, and he's used to jumping to holds, cutting his feet and taking the swing. He's strong enough to climb with this style, but he's also heavier, which means he has to climb faster so as to not get too tired. 'Skip the bad holds, go for the good ones,' he says in the climbing movie Dosage 3. Both Dave and Chris have developed and perfected their styles over time. Both could have trained to become more like the other, and to a certain extent they probably have, but Dave's style wouldn't work for Chris, and vice versa. They both have a technique which is perfectly suited to their body.

Progressing your technique involves more than just knowing about the different fundamentals and techniques. You need to stay curious to find the optimal solution for you, learning from the feedback your body and your peers give you, and be conscious not just of if you get up a climb, but how you got up it.



